

In-situ IR measurment with ReactIR 701L in GL14 flask under inert conditions

Date: 2025-06-02
Tags: ATR-IR AE IR in-situ in-situ IR in progress
Category: Protocol
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Goal

In-situ IR measurment of homogeneous liquid systems under inert conditions

Prerequisites and preparation

Information - ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe

Computer with software + display, mouse and keyboard

Desired reactor (with GL14 connection)

Desired system

Idea what you want to observe and hwo to conduct the specific experiment to achieve that

BOLA fitting - 6.35 mm

Funnel for liquid nitrogen

Safety

Do not bend glas fiber cable with force

Do not make tight turn (like less then 15 cm)

Work in an analytic fumehood - no unnecessary stuff is allowed in the fumehood

General remarks

Can measure down to approx. 2 mM concnetration, with medium IR absorber

Do not touch IR Probe during experiment, espacially do not cause vibrations --> will yield offset in data

When chainging falsk to measure in make sure to place the probe in exactly the same position as before, otherwise offset is possible

Steps

Step number	Step description	Pictures
-1. Before starting	Get the device from ZAF 107 and move it to where you want to use it (before that infrom responsible person for device)	

0. Starting the device	Connect the Computer, ReactIR and display to a power source. Connect the ReactIR, the display and the keyboard to the Computer Connect the Probe to the ReactIR Start the Computer, Display and ReactIR	
1. Before the experiment	Add liquid nitrogen to the ReactIR (approx. 1.5 L) until it is filled Wait for at least 4 h for equilibration of the system	
2. Start software	Click on the icon iC IR 7.1 on the desktop	icon.PNG
3. Setup experiment	You get the view shown in start. Click on Quick Start	Start.PNG
3a. errors	No errors should occur. If error occurs typically wait longer and also errors can occur directly after adding new liquid N ₂ (so if there are errors wait like 30 min and check again) Only error to continue working with is when the IR Probe Temperature is not found	errors.PNG
3b. Change of further experiment parameters	If desired navigate to tab File, click on instrument and adjust desired parameters: you can change: resolution and wavenumber range, other should not be changed without knowing what you want to do	Change parameters.PNG
4. Save experiment	Windown opens: Give name and choose folder to save file Set experiment duration (not sure if it can be changed later on) And select sample Interval: can be changed later, till 3 min number of spectra per average increases till 256, afterwards number of spectra per measurement stays constant and only waiting time between 2 measurements increases When done click on create	file-saving.PNG

5. Measure background	Typically you are asked to measure the background, other wise you can the window of step 6 will open and you can manually collect the backgorund. The background should be measured in air, the probe should be in exactly the same position as during the experiment <table border="1"> <tr> <td data-bbox="395 297 595 705">a. Clean probe</td><td data-bbox="595 297 876 705">Clean probe with appropriate solvent (all organic solvents and water are fine, do not use strong acids or bases, could corrod steel) When Test status is clean click on next</td><td data-bbox="876 297 1120 705">clean.PNG</td></tr> <tr> <td data-bbox="395 705 595 869">b. Collect background</td><td data-bbox="595 705 876 869">Choose appropriate 256 scans and click on collect background</td><td data-bbox="876 705 1120 869">background.PNG</td></tr> <tr> <td data-bbox="395 869 595 1066">c. done</td><td data-bbox="595 869 876 1066">click on save background or something similar, its clear what you should click</td><td data-bbox="876 869 1120 1066"></td></tr> </table>	a. Clean probe	Clean probe with appropriate solvent (all organic solvents and water are fine, do not use strong acids or bases, could corrod steel) When Test status is clean click on next	clean.PNG	b. Collect background	Choose appropriate 256 scans and click on collect background	background.PNG	c. done	click on save background or something similar, its clear what you should click		
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b. Collect background	Choose appropriate 256 scans and click on collect background	background.PNG									
c. done	click on save background or something similar, its clear what you should click										
6. Start experiment	Window opens Press arrow to start measurment You can click throught the different visulaizations and so and zoom in and out to them, you can also click through the different tabs, even so by now I did not find anything usefull there	interface.PNG									
7. During experiment	Do the experiment you want to do If wanted: Pause experiment and adjust interval, afterwards press start again If flask to measure in needs to be changes: keep IR probe in same position and pause experiement during this If finished: click on stop experiment, than experiment cannot be resumed and all data is automatically saved in the proprietary file format										

8. Clean Probe and measure background	Analog to 5 you will be asked to measure the background. Do that		
	a. Clean probe	Clean probe with appropriate solvent (all organic solvents and water are fine, do not use strong acids or bases, could corrode steel) When Test status is clean click on next	clean.PNG
	b. Collect background	Choose appropriate number of scans (depends if you want to continue measurement, then choose 256 scans, otherwise lower number is also fine) and click on collect background	background.PNG
	c. done	click on save background or something similar, its clear what you should click	
8. Export data	Click on the tab file --> Export --> Multi-spectrum file Choose appropriate name and folder Format: CSV, Data: raw Click on export		
9. Finish	You can collect data on USB stick and shutdown the program. Keep IR device plugged in for min 2 more hours, to be sure		
10. Finished with experiment series	If no experiments planned in the next days disassemble everything and put it out of your / the analytic fumehood and store it somewhere around the lab If no experiments planned in the next month: return everything back to ZAF 107 where you took it from		

Linked resources

Equipment - [EPR, CEEC I lab K002](#)

Equipment - [H2 UniAmp Sensor - Low range - 2.1 x 80 mm needle](#)

Equipment - [H2 UniAmp Single Channel System](#)

Equipment - [ReactIR 701L with DST 6.35 x 1.5 m x 305 DiComp probe](#)

Information - [Everything about Usage of Irradiation Set-Up](#)

Protocol - [EPR measurement on EPR, CEEC I lab K002](#)

Protocol - [Getting hydrogen from hydrogen bottle in CEEC II E014](#)

Protocol - [Gas phase calibration of H2 UniAmp sensor](#)

Protocol - [Hydrogen measurement with H2 UniAmp sensor \(liquid or gas phase continuous measurement\)](#)

Protocol - [Liquid phase calibration of H2 UniAmp sensor](#)

Attached files

Change-parameters.PNG

sha256: d6095621625fea06a591459ccd47548b56f56a2871713ce84dbf9ee5d22159a8



errors.PNG

sha256: dcfba4d4864ea55cab840ea5635b089bac558c3ba26b724b8354b4d2b72ed61e



background.PNG

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clean.PNG

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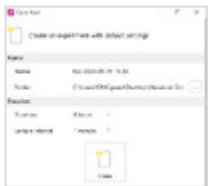
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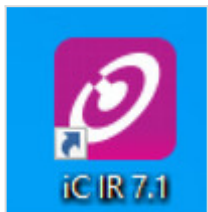
file-saving.PNG

sha256: 5b2de46ade980c6a9790209edad2123250ca01b6de9ac3cdd0854bfdd06b0839



icon.PNG

sha256: b7772498de323499c25d6380f0efe1f81a34489c57de5839d00a0dc12a13e6e5



Start.PNG

sha256: 12d8c082b8fc82e89ac38a27edf51f6b48a32a6a590f51685fb2638380207529



Unique eLabID: 20250602-9e170d4328e742f9bf8f348d6c63fce5ca1d48f9
Link: <https://elab.water-splitting.org/database.php?mode=view&id=235>